

probability.md

Generalized Arc Consistency on a Constraint Network

This task involves applying the Generalized Arc Consistency (GAC) algorithm to the constraint network $\gamma = (V, D, C)$:

Problem Statement

- **Variables:**

$$V = \{a, b, c, d\}$$

- **Domains:**

$$\text{For all } v \in V: D_v = \{1, 2, 3, 4, 5\}$$

- **Constraints:**

- $a + 2 < d$
- $b \times d < 6$
- $a + c < 6$

Your task: **Run the Generalized Arc Consistency algorithm** on this constraint network. After enforcing GAC, what are the possible values (domains) for each variable?

Solution

Let's enforce GAC on each constraint step by step.

1. Constraint: $a + 2 < d$

For each possible $a \in \{1, 2, 3, 4, 5\}$, determine possible $d \in \{1, 2, 3, 4, 5\}$ such that $a + 2 < d$.

- $a = 1 \Rightarrow 1 + 2 = 3 < d \Rightarrow d \in \{4, 5\}$
- $a = 2 \Rightarrow 2 + 2 = 4 < d \Rightarrow d = 5$
- $a = 3 \Rightarrow 3 + 2 = 5 < d \Rightarrow d > 5 \Rightarrow d = \emptyset$ (no solution)

- $a = 4 \Rightarrow 4+2=6 < d \Rightarrow d > 6 \Rightarrow d = \emptyset$
- $a = 5 \Rightarrow 5+2=7 < d \Rightarrow d > 7 \Rightarrow d = \emptyset$

So $a \in \{1, 2\}$. Corresponding feasible d (due to this constraint):

- For $a=1 \Rightarrow d \in \{4, 5\}$
- For $a=2 \Rightarrow d=5$

Therefore, for this constraint, $d \in \{4, 5\}$.

2. Constraint: $b \times d < 6$

Original: $b, d \in \{1, 2, 3, 4, 5\}$. Update d to $\{4, 5\}$ (from above):

Check for each $b \in \{1, 2, 3, 4, 5\}$ and $d \in \{4, 5\}$:

- $d=4$:
 - $b=1: 1 \times 4 = 4 < 6$ ✓
 - $b=2: 2 \times 4 = 8$ ✗
 - $b=3: 3 \times 4 = 12$ ✗
 - $b=4: 4 \times 4 = 16$ ✗
 - $b=5: 5 \times 4 = 20$ ✗
- $d=5$:
 - $b=1: 1 \times 5 = 5 < 6$ ✓
 - $b=2: 2 \times 5 = 10$ ✗
 - $b=3: 3 \times 5 = 15$ ✗
 - $b=4: 4 \times 5 = 20$ ✗
 - $b=5: 5 \times 5 = 25$ ✗

So, for both $d=4$ and $d=5$, only $b=1$ is possible.

Thus, $b = \{1\}$

3. Constraint: $a + c < 6$

Recall $a = \{1, 2\}$

- For $a=1$:
 - $1+c < 6 \Rightarrow c < 5 \Rightarrow c \in \{1, 2, 3, 4\}$
- For $a=2$:
 - $2+c < 6 \Rightarrow c < 4 \Rightarrow c \in \{1, 2, 3\}$

So, possible c are those that work for at least one a :

- For $a=1$: $c \in \{1,2,3,4\}$
- For $a=2$: $c \in \{1,2,3\}$

But GAC means for every remaining c , there must exist a compatible a as well.

- $c=1$: ($a=1$ or $a=2$)
- $c=2$: ($a=1$ or $a=2$)
- $c=3$: ($a=1$ or $a=2$)
- $c=4$: (only $a=1$)

So $c \in \{1,2,3,4\}$

But let's check reversibility: for every a , there must be a c :

- $a=1$: $c \in \{1,2,3,4\}$ (all fine)
- $a=2$: $c \in \{1,2,3\}$ (4 does not work)

So, update:

- $a=1$: compatible with $c \in \{1,2,3,4\}$
- $a=2$: compatible with $c \in \{1,2,3\}$ But $c=4$ is only valid if $a=1$, and since $a=1$ is still in the domain, that's acceptable.

So final domains:

- $a \in \{1,2\}$
- $b \in \{1\}$
- $c \in \{1,2,3,4\}$
- $d \in \{4,5\}$

Final Domains after GAC:

- $a = \{1, 2\}$
- $b = \{1\}$
- $c = \{1, 2, 3, 4\}$
- $d = \{4, 5\}$

README: Generalized Arc Consistency

Example

This document presents a constraint satisfaction problem and guides you through running the Generalized Arc Consistency (GAC) algorithm to prune variable domains. At the end, both the to-do list for arcs and the pruned domains are provided.

Problem Definition

Constraint Network:

$$\gamma = (V, D, C)$$

- **Variables:** $V = \{a, b, c, d, e\}$
- **Domains:** For all $v \in V$: $D_v = \{1, 2, 3, 4, 5, 6\}$
- **Constraints:**
 - $a + d > 8$
 - $4e - 2b < 3$
 - $c + 2 > a$
 - $b < d$

Task

1. **Run the Generalized Arc Consistency (GAC) algorithm** on the above constraint network.
2. **Select the valid To-do Arcs** for the first iteration of the GAC algorithm.
3. **List the pruned domains** of all variables after enforcing GAC.

Solution

Step 1: Represent each constraint as arcs

The binary constraints induce the following arcs (in both directions for a binary constraint):

- $a + d > 8 \Rightarrow (a, d), (d, a)$
- $b < d \Rightarrow (b, d), (d, b)$

- $c + 2 > a \Rightarrow (c, a), (a, c)$
- $4e - 2b < 3 \Rightarrow (e, b), (b, e)$

Thus, the initial To-Do list of arcs (all arcs affected by some constraint):

- $(a, d), (d, a)$
- $(b, d), (d, b)$
- $(c, a), (a, c)$
- $(e, b), (b, e)$

Step 2: Valid To-Do Arcs for First Iteration

From the options given, the correct set for the initial to-do list is:

$(a, d), (d, a), (b, d), (d, b), (a, c), (c, a), (e, b), (b, e)$

This matches the set in the option:

$(e, b), (d, b), (d, a), (c, a), (b, e), (b, d), (a, d), (a, c)$

Step 3: Run GAC and Prune Domains

1. Constraint: $a + d > 8$

Possible values:

Both a and d range from 1 to 6.

- For $a = 3$, d must be $> 5 \Rightarrow$ impossible, since $\max d = 6$.
- Let's brute-force all possibilities:

a	d	$a+d$	$a+d>8?$
3	6	9	YES
4	5	9	YES
4	6	10	YES
5	4	9	YES
5	5	10	YES
5	6	11	YES

a	d	a+d	a+d>8?
6	3	9	YES
6	...	...	YES if $d \geq 3$

It turns out, the pairs for which $a + d > 8$ are:

- (3,6)
- (4,5), (4,6)
- (5,4), (5,5), (5,6)
- (6,3), (6,4), (6,5), (6,6)

So possible a values: 3,4,5,6

Possible d values: 3,4,5,6

But wait: let's check which values of a and d actually occur in the allowed pairs.

Set of possible a:

- a = 3 occurs with d = 6
- a = 4 with d = 5,6
- a = 5 with d = 4,5,6
- a = 6 with d = 3,4,5,6

Set of possible d:

- d=3 with a=6
- d=4 with a=5,6
- d=5 with a=4,5,6
- d=6 with a=3,4,5,6

Therefore:

- $a \in \{3,4,5,6\}$
- $d \in \{3,4,5,6\}$

2. Constraint: $4e - 2b < 3$

That is, $4e - 2b < 3 \Rightarrow 2e - b < 1.5$

Let's just test all values for e and b from 1 to 6.

Try e = 1:

- $b = \text{any} \Rightarrow 4 \times 1 - 2b < 3 \Rightarrow 4 - 2b < 3 \Rightarrow -2b < -1 \Rightarrow b > 0.5 \Rightarrow b \in \{1,2,3,4,5,6\}$ But let's check further since b is integer.

Test for $e = 1$: $4 \times 1 - 2b < 3 \Rightarrow 4 - 2b < 3 \Rightarrow -2b < -1 \Rightarrow b > 0.5$

So, $b \in \{1,2,3,4,5,6\}$

$e = 2$: $4 \times 2 - 2b < 3 \Rightarrow 8 - 2b < 3 \Rightarrow -2b < -5 \Rightarrow b > 2.5 \Rightarrow b \in \{3,4,5,6\}$

$e = 3$: $4 \times 3 - 2b < 3 \Rightarrow 12 - 2b < 3 \Rightarrow -2b < -9 \Rightarrow b > 4.5 \Rightarrow b \in \{5,6\}$

$e = 4$: $16 - 2b < 3 \Rightarrow -2b < -13 \Rightarrow b > 6.5 \Rightarrow b \in \{\}$ (empty)

$e = 5$: $20 - 2b < 3 \Rightarrow -2b < -17 \Rightarrow b > 8.5 \Rightarrow b \in \{\}$

$e = 6$: $24 - 2b < 3 \Rightarrow -2b < -21 \Rightarrow b > 10.5 \Rightarrow b \in \{\}$

Therefore,

- $e \in \{1,2,3\}$
- For $e = 1 \Rightarrow b \in \{1,2,3,4,5,6\}$
- For $e = 2 \Rightarrow b \in \{3,4,5,6\}$
- For $e = 3 \Rightarrow b \in \{5,6\}$

But if e is restricted, the possible domains are:

- $e \in \{1,2,3\}$
- $b \in \{1,2,3,4,5,6\}$ Now, considering only possible e 's, must prune b further.

From the above:

- $b \in \{1,2,3,4,5,6\}$: compatible with $e=1$
- $b \in \{3,4,5,6\}$: compatible with $e=2$
- $b \in \{5,6\}$: compatible with $e=3$

All this will be further pruned by other constraints (see next).

3. Constraint: $c + 2 > a$

For pruned $a \in \{3,4,5,6\}$, so $c > a - 2$

Let's see for what values of c ($1,2,3,4,5,6$) and $a \in \{3,4,5,6\}$, $c + 2 > a \Rightarrow c > a - 2$

For $a = 3$: $c > 1 \Rightarrow c \in \{2,3,4,5,6\}$ $a = 4$: $c > 2 \Rightarrow c \in \{3,4,5,6\}$ $a = 5$: $c > 3 \Rightarrow c \in \{4,5,6\}$ $a = 6$: $c > 4 \Rightarrow c \in \{5,6\}$

So union over $a \in \{3,4,5,6\} = c \in \{2,3,4,5,6\}$

But, for a particular value of c , are there any compatible a values left with all the constraints? Let's keep $c \in \{2,3,4,5,6\}$ for now.

4. Constraint: $b < d$

$d \in \{3,4,5,6\}$; $b \in ?$

Let's consider all candidate b 's from earlier ($\{1,2,3,4,5,6\}$). For d values $\{3,4,5,6\}$, b can be at most 5.

But with previous constraints (like $4e - 2b < 3$) and domain restrictions, finally feasible values are:

- b must be less than at least one value of d . So possible $b \in \{1,2,3,4,5\}$

But factoring in all constraints, and cross-referencing with e domain:

From $4e - 2b < 3$, we had

- If $e = 1 \rightarrow b \in \{1,2,3,4,5,6\}$
- If $e = 2 \rightarrow b \in \{3,4,5,6\}$
- If $e = 3 \rightarrow b \in \{5,6\}$

But b cannot be above 5, since $d \in \{3,4,5,6\} \Rightarrow b < d \Rightarrow b \in \{1,2,3,4,5\}$

For $b = 6$ to be valid, need $d = 6 + \text{something} \Rightarrow d = 6$ but $b < 6$ fails for $b = 6$. So $b = 6$ excluded.

So finally $b \in \{1,2,3,4,5\}$

$e = 1$ is compatible with $b \in \{1,2,3,4,5\}$, $e = 2$ with $b \in \{3,4,5\}$, $e = 3$ with $b \in \{5\}$, else e out of domain.

So, all considered, final feasible values:

- $a \in \{3,4,5,6\}$
- $b \in \{1,2,3,4,5\}$
- $c \in \{2,3,4,5,6\}$
- $d \in \{3,4,5,6\}$
- $e \in \{1,2,3\}$

Summary Table

Variable	Final Domain
a	3, 4, 5, 6
b	1, 2, 3, 4, 5
c	2, 3, 4, 5, 6
d	3, 4, 5, 6
e	1, 2, 3

Answers

1. Valid To-Do Arcs for the first iteration of GAC:

(e, b), (d, b), (d, a), (c, a), (b, e), (b, d), (a, d), (a, c)

2. Domains after GAC:

- $a \in \{3, 4, 5, 6\}$
- $b \in \{1, 2, 3, 4, 5\}$
- $c \in \{2, 3, 4, 5, 6\}$
- $d \in \{3, 4, 5, 6\}$
- $e \in \{1, 2, 3\}$

Generalized Arc Consistency Example: README and Solution

Constraint Network

Variables:

$V = \{a, b, c, d\}$

Domains:

$D_v = \{1, 2, 3\}$ for all v in V

Constraints:

- $b = a$
- $b > c$
- $a \neq c$
- $c \neq d$
- $d \leq a$

Task

Run the **Generalized Arc Consistency (GAC) algorithm** on this network.

Determine the resulting domains for each variable.

Solution

Let's apply GAC step by step:

1. Initial Domains

- $a \in \{1, 2, 3\}$
- $b \in \{1, 2, 3\}$
- $c \in \{1, 2, 3\}$
- $d \in \{1, 2, 3\}$

2. Constraint Propagation

Let's propagate each constraint in turn, applying pruning as needed.

a. $b = a$

- For every (a, b) pair, must have $a = b$.
So, a and b must always be equal.

Therefore, after this constraint:

- a, b domains must be equal at each pruning step.

b. $b > c$

- For possible $b \in \{1,2,3\}$ and $c \in \{1,2,3\}$, only consider pairs where $b > c$.
- The valid pairs:
 - $b = 2, c = 1$
 - $b = 3, c = 1$
 - $b = 3, c = 2$
- Therefore, **possible values for b: 2, 3** (since no $b = 1 > c$ for $c \in \{1,2,3\}$), and for c: 1, 2 (since only $c = 1$ or 2 can appear in $\{b > c\}$).

But **remember $b = a$** , so $a \in \text{domain}(b) \Rightarrow a \in \{2, 3\}$.

After this step:

- $a \in \{2, 3\}$
- $b \in \{2, 3\}$
- $c \in \{1, 2\}$
- $d \in \{1, 2, 3\}$

c. $a \neq c$

- $a \in \{2,3\}$
- $c \in \{1,2\}$
- For $a = 2$, c can be 1 (since $a \neq c$)
- For $a = 3$, c can be 1 or 2 (since $a \neq c$ always holds: $3 \neq 1, 3 \neq 2$)

So,

- For $c = 2$: only compatible with $a = 3$

d. $c \neq d$

- $c \in \{1, 2\}$
- $d \in \{1, 2, 3\}$

So for each c :

- $c = 1$: $d \in \{2, 3\}$
- $c = 2$: $d \in \{1, 3\}$

e. $d \leq a$

$$a \in \{2, 3\} \quad d \in \{1, 2, 3\}$$

$$\text{For } a = 2: d \in \{1, 2\}$$

$$\text{For } a = 3: d \in \{1, 2, 3\}$$

Now, let's express all possible consistent tuples and prune incompatible values.

Let's enumerate all possible values for (a, b, c, d) consistent with all constraints.

Remember:

- $b = a$
- $b > c$
- $a \neq c$
- $c \neq d$
- $d \leq a$

Loop over $a \in \{2, 3\}$, $b = a$, $c \in \{1, 2\}$, $d \in$ possible values.

a = 2, b = 2:

- $c \in \{1\}$
 - $b > c \rightarrow 2 > 1 \checkmark$
 - $a \neq c: 2 \neq 1 \checkmark$
 - $c \neq d \rightarrow d \neq 1$
 - $d \leq a \rightarrow d \in \{1, 2\}$
 - So: $d = 2$ (since $d \neq 1$ by $c \neq d$).
 - So possible tuple: (a=2, b=2, c=1, d=2)
- $c = 2:$
 - $b > c \rightarrow 2 > 2 \times$ (no, must be $b > c$)
 - skip

a = 3, b = 3:

- $c = 1:$
 - $b > c \rightarrow 3 > 1 \checkmark$
 - $a \neq c: 3 \neq 1 \checkmark$
 - $c \neq d: d \neq 1$
 - $d \leq a \rightarrow d \in \{1, 2, 3\}$
 - $d \in \{2, 3\}$

- So, possible tuples:
 - $(a=3, b=3, c=1, d=2)$
 - $(a=3, b=3, c=1, d=3)$
- $c = 2$:
 - $b > c \rightarrow 3 > 2 \checkmark$
 - $a \neq c: 3 \neq 2 \checkmark$
 - $c \neq d \rightarrow d \neq 2$
 - $d \leq a \rightarrow d \in \{1, 2, 3\}$
 - $d \in \{1, 3\}$
 - So, possible tuples:
 - $(a=3, b=3, c=2, d=1)$
 - $(a=3, b=3, c=2, d=3)$

3. Final Pruned Domains

From the possible tuples above, extract values that appear for each variable:

- a : appears as 2, 3
- b : equal to $a \Rightarrow \{2, 3\}$
- c : 1, 2
- d : 1, 2, 3 (let's see which d 's occur: 2, 2, 3, 1, 3 $\rightarrow$ so 1, 2, 3; but check which values for which a)

Mapping:

- $(a=2, b=2, c=1, d=2)$
- $(a=3, b=3, c=1, d=2)$
- $(a=3, b=3, c=1, d=3)$
- $(a=3, b=3, c=2, d=1)$
- $(a=3, b=3, c=2, d=3)$

So the final domains are:

- $a \in \{2, 3\}$
- $b \in \{2, 3\}$
- $c \in \{1, 2\}$
- $d \in \{1, 2, 3\}$

Summary Table

Variable	Domain
a	{2, 3}
b	{2, 3}
c	{1, 2}
d	{1, 2, 3}

README: Variable Elimination Algorithm for CSPs

Question:

In the variable elimination algorithm to solve CSPs, which of the following is correct?

Select one:

a.

We construct a table for each constraint, and at each step the algorithm removes a constraint by combining all its variables.

b.

We construct a table for each constraint, and at each step the algorithm removes a variable by combining all its constraints.

c.

We construct a table for each variable, and at each step the algorithm removes a constraint by combining all its variables.

d.

We construct a table for each variable, and at each step the algorithm removes a variable by combining all its constraints.

Solution

The correct answer is:

b. We construct a table for each constraint, and at each step the algorithm removes a variable by combining all its constraints.

Explanation:

In the variable elimination algorithm for CSPs (Constraint Satisfaction Problems):

- Initially, a "factor" (table of allowed values) is constructed for each constraint.
- At each step, the algorithm selects a variable to eliminate. All the tables ("factors") involving that variable are combined into a new table (by joining and summing/eliminating over the selected variable).
- This process continues — eliminating one variable at a time — until all variables are removed or a solution is produced.

So the process is not about removing constraints, but about eliminating variables by combining the relevant constraint tables.

Answer: b

Analyzing Independence in Categorical Data: Positive Reviews & Discounts

Consider the following scenario:

There are **10,000 reviews** for a restaurant, each annotated with three boolean variables:

- **Positive** (was the review positive?)
- **Discount** (was a discount offered?)
- **Long** (was the review long or short?)

The following table (see image below) displays the counts for each combination of these variables:

	<i>Discount</i>			
	<i>yes</i>		<i>no</i>	
<i>Positive</i>	<i>Long</i>		<i>Long</i>	
	<i>yes</i>	<i>no</i>	<i>yes</i>	<i>no</i>
<i>yes</i>	130	170	1000	1700
<i>no</i>	500	200	4000	2300

Dependence for exercise

Your tasks:

1. Are “Positive” and “Discount” independent?

Determine whether the variables **Positive** and **Discount** are independent of each other.

2. Joint and Marginal Distributions

Fill in the joint probabilities for the following outcomes (all as probabilities, not counts):

- **P(Positive=Yes, Discount=Yes):**
- **P(Positive=Yes, Discount=No):**
- **P(Positive=No, Discount=Yes):**
- **P(Positive=No, Discount=No):**

Also, compute the marginal probabilities:

- **P(Positive=Yes):**
- **P(Positive=No):**
- **P(Discount=Yes):**
- **P(Discount=No):**

3. Independence Check

Based on your calculations above, do **Positive** and **Discount** appear to be independent?

State your reasoning and show your calculations or explanation.

Here is the answer formatted for a README file:

Restaurant Review Independence Analysis



Data Summary

From a dataset of **10,000 restaurant reviews**, we have three boolean variables:

Variable	Description
Positive	Whether the review is positive
Discount	Whether a discount was mentioned
Long	Whether the review is long

Data Table:

	Discount = yes	Discount = no
	Long = yes	Long = no
Positive = yes	130	170
Positive = no	500	200



Step 1 – Marginalize over Long

We sum over Long to get joint counts of (Positive, Discount) .

- **Positive = yes, Discount = yes**
(130 + 170 = 300)
- **Positive = yes, Discount = no**
(1000 + 1700 = 2700)

- **Positive = no, Discount = yes**
(500 + 200 = 700)
- **Positive = no, Discount = no**
(4000 + 2300 = 6300)

Check total:

(300 + 2700 + 700 + 6300 = 10000) ✓



Step 2 – Joint & Marginal Probabilities

Joint distribution (P(Positive, Discount)) (divide by 10000):

(P)	Discount = yes	Discount = no	Marginal (P(Positive))
Positive = yes	0.0300	0.2700	0.3000
Positive = no	0.0700	0.6300	0.7000
Marginal (P(Discount))	0.1000	0.9000	



Step 3 – Test for Independence

Two variables are independent if:

[$P(\text{Positive}, \text{Discount}) = P(\text{Positive}) \times P(\text{Discount})$]

- $(P(\text{Positive}=\text{yes}) \times P(\text{Discount}=\text{yes}) = 0.3 \times 0.1 = 0.03)$
→ Matches joint (0.03) ✓
- $(P(\text{Positive}=\text{yes}) \times P(\text{Discount}=\text{no}) = 0.3 \times 0.9 = 0.27)$
→ Matches joint (0.27) ✓
- $(P(\text{Positive}=\text{no}) \times P(\text{Discount}=\text{yes}) = 0.7 \times 0.1 = 0.07)$
→ Matches joint (0.07) ✓
- $(P(\text{Positive}=\text{no}) \times P(\text{Discount}=\text{no}) = 0.7 \times 0.9 = 0.63)$
→ Matches joint (0.63) ✓

All match exactly.



Conclusion

Positive and Discount are independent in this dataset.



Summary Table (Joint Distribution)

(P(Positive, Discount))	Discount = yes	Discount = no
Positive = yes	0.0300	0.2700
Positive = no	0.0700	0.6300

Marginals:

- $(P(\text{Positive}=\text{yes}) = 0.3000)$
- $(P(\text{Positive}=\text{no}) = 0.7000)$
- $(P(\text{Discount}=\text{yes}) = 0.1000)$
- $(P(\text{Discount}=\text{no}) = 0.9000)$